

Xiaoxia (Shirley) Wu

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in [xiaoxiawu1](#)

Google Scholar

Summary

Principal Scientist with 10+ years of experience building production-scale LLM systems across training, serving, and post-training optimization. Led production-scale inference and model compression initiatives at Together AI and Microsoft, spanning speculative decoding, KV-cache quantization, FP6 serving, and RL-driven post-training. Published 30+ papers with 7,300+ citations (h-index 24), with research deployed in modern LLM serving stacks including DeepSpeed, vLLM, SGLang, and TensorRT-LLM.

Experience

- Jun. 2026 – **Member of Technical Staff, Mirendil** (\$200M seed funding; \$1B valuation), SF, CA
Present
 - Building scalable RL-driven post-training infrastructure for large language models.
- Jul. 2024 – **Principal Scientist, Together AI**, Bellevue, WA
- Jun. 2026
 - Led **Aurora** and **ATLAS**, unified training-serving systems for speculative decoding that reduce training-serving mismatch through continuous online adaptation from live inference traffic.
 - Built and scaled production speculator training and distillation pipelines spanning supervised fine-tuning, distillation, and RL-driven post-training.
 - Drove full-stack inference efficiency across quantization formats including FP8, FP6, NVFP4, INT4, INT2, ternary, and binary, with deployment in vLLM, SGLang, and TensorRT-LLM.
 - Published technical blogs on [Aurora](#), [ATLAS](#), and [DeepSeek-R1 inference on NVIDIA HGX B200](#).
- Nov. 2021 – **Senior Researcher, DeepSpeed Team, Microsoft**, Redmond, WA
- Jun. 2024
 - Led cross-functional quantization efforts spanning algorithms, kernels, and serving integration, including [DeepSpeed-FP6](#), [ZeroQuant-FP](#), [ZeroQuant-HERO](#), and [ZeroQuant](#), achieving significant inference speedups and memory reductions for large language models.
 - Co-led development of [DeepSpeed-Chat](#) and [DeepSpeed-VisualChat](#), enabling affordable RLHF training of ChatGPT-like models.
 - Published high-impact work including *Extreme Compression for Pre-trained Transformers Made Simple* ([NeurIPS 2022 Oral](#)).
- Jan. 2021 – **Postdoctoral Researcher, University of Chicago**, Host: Rebecca Willett
- Oct. 2021
 - Developed algorithms for differentially private empirical risk minimization.
 - Conducted theoretical analysis on privacy-utility tradeoffs in machine learning models.
- May 2020 – **Research Intern, Google**, Mountain View, CA, Mentor: Behnam Neyshabur
- Aug. 2020
 - Researched curriculum learning for CIFAR10 and ImageNet.
 - Demonstrated that curriculum learning provides no consistent benefit over standard training; results published as [ICLR 2021 oral](#).
- Aug. 2017 – **Research Intern, Meta AI Research**, New York, NY, Mentor: Léon Bottou
- Oct. 2018
 - Implemented and benchmarked state-of-the-art optimization algorithms in PyTorch.
 - Analyzed optimization of layer and weight normalization; work presented as [ICML 2019 oral](#).

Publications and Preprints (selected)

Google Scholar: 7,300 citations, h-index 24, i10-index 30 (as of July 2026).

LLM Systems, Reasoning, and Speculative Decoding

- [1] J. Wang, F. Bie, J. Li, Z. Zhou, Z. Shao, Y. Wang, Y. Liu, et al, **X. Wu (project lead)**. *When RL Meets Adaptive Speculative Training*. [arXiv:2602.06932](#) [ICML](#) (2026)
- [2] Z. Zhou, F. Bie, Z. Chen, et al, **X. Wu (project lead)**, S.L. Song. *CARE: Covariance-Aware and Rank-Enhanced Decomposition for Enabling Multi-Head Latent Attention*. [ICLR](#) (2026).
- [3] Fengxiang Bie, Yuqing Jian, et al, **X. Wu**, Tianyi Zhang. *Osprey: Target-agnostic Pre-training Makes Stronger Drafters in Speculative Decoding*. *Under Review* (2026).
- [4] H. Singh, X. Li, K. Sareen, M. Maheswaran, S. Tan, **X. Wu**, et al. *V1: Unifying Generation and Self-Verification for Parallel Reasoners*. [arXiv:2603.04304](#) (2026).

- [5] Z. Shao, V. Srivatsa, Q. Wu, A. Ariyak, **X. Wu**, et al. *Beat the Long Tail: Distribution-Aware Speculative Decoding for RL Training*. [arXiv:2511.13841](#) [MLSys](#) (2026).
- [6] Z. Zhang, **X. Wu (project lead)**, et al. *Understanding and Steering the Cognitive Behaviors of Reasoning Models at Test-Time*. [arXiv:2512.24574](#) (2025).

Model Compression

- [1] Z. Zhou, et al., **X. Wu (project lead)**. *OSCAR: Offline Spectral Covariance-Aware Rotation for 2-bit KV Cache Quantization*. [arXiv:2605.17757](#).
- [2] J. Jia, et al., **X. Wu (project lead)**. *Saw-int4: System-Aware 4-Bit KV-Cache Quantization for Real-World LLM Serving*. [arXiv:2604.19157](#).
- [3] T. Gupta, H. Prairie, **X. Wu**, et al. *Search Your Block Floating Point Scales!* [MLSys](#) 2026.
- [4] H. Xia, **X. Wu (project lead)**, et al. *Kitty: Accurate and Efficient 2-bit KV Cache Quantization*. [arXiv:2511.18643](#). [MLSys](#) 2026.
- [5] H. Xia, Z. Zheng, **X. Wu**, et al. *FP6-LLM: Efficiently Serving LLMs through FP6-Centric Algorithm-System Co-Design*. [arXiv:2401.14112](#), 2024.
- [6] **X. Wu**, H. Xia, S. Youn, Z. Zheng, et al. *ZeroQuant(4+2): Redefining LLM Quantization with an FP6 Strategy*. [arXiv:2312.08583](#) (2023).
- [7] **X. Wu***, Z. Yao*, et al. *ZeroQuant-FP: W4A8 Post-Training Quantization Using Floating-Point Formats*. [NeurIPS ENLSP](#) (2023).
- [8] G. Wang, H. Qin, S.A. Jacobs, **X. Wu**, et al. *ZeRO++: Extremely Efficient Collective Communication for Large Model Training*. [ICLR](#) (2024).
- [9] Z. Yao, **X. Wu**, C. Li, et al. *ZeroQuant-V2: Exploring Post-Training Quantization in LLMs*. [AAAI](#) (2024).
- [10] **X. Wu**, C. Li, et al. *Understanding INT4 Quantization for Transformer Models*. [ICML](#) (2023).
- [11] **X. Wu**, et al. *Extreme Compression for Pre-trained Transformers Made Simple*. [NeurIPS](#) (2022), [oral](#).

Optimization and Theory

- [1] **X. Wu**, E. Dyer, B. Neyshabur. *When Do Curricula Work?* [ICLR](#) (2021), [oral](#).
- [2] **X. Wu**, Y. Xie, et al. *Adaloss: A Computationally-Efficient Adaptive Gradient Method*. [AAAI](#) (2022).
- [3] **X. Wu***, et al. *Implicit Regularisation and Convergence for Weight Normalisation*. [NeurIPS](#) (2020).
- [4] R. Ward*, **X. Wu***, L. Bottou. *Adagrad Stepsizes: Sharp Convergence over Nonconvex Landscapes*. [ICML](#) (2019), [oral](#).

NLP and Multimodal

- [1] Z. Yao, **X. Wu**, C. Li, M. Zhang, H. Qin, et al. *DeepSpeed-VisualChat: Multi-Round Multi-Image Interleave Chat*. [arXiv:2309.14327](#) (2023).
- [2] Z. Yao, R.Y. Aminabadi, O. Ruwase, S. Rajbhandari, **X. Wu**, et al. *DeepSpeed-Chat: Affordable RLHF Training of ChatGPT-like Models*. [arXiv:2308.01320](#) (2023).
- [3] C. Li, Z. Yao, **X. Wu**, M. Zhang, Y. He. *DeepSpeed Data Efficiency: Improving Deep Learning Model Quality*. [AAAI](#) (2024).

Honours and Awards

- 2020 Frank Gerth III Dissertation Award (top dissertation in Mathematics, UT Austin)
- 2019 ICML & NeurIPS Travel Awards
- 2018, 2019 Professional Development Award, UT Austin
- 2018 Graduate School Fellowship, UT Austin
- 2012 Scotland Saltire Scholarship, University of Edinburgh

Professional Service & Mentorship

- Journal Reviewer: *Journal of Machine Learning Research*
- Conference Reviewer: NeurIPS, ICML, ICLR, AISTATS, MSML, WiML
- Mentor & Volunteer Teacher: Sanger Center & Directed Reading Program, UT Austin

Education

- Aug. 2014 – **Ph.D. in Machine Learning**, *The University of Texas at Austin*
- Dec. 2020
 - Advisors: Rachel Ward (supervisor) & Léon Bottou (co-supervisor)
 - Thesis (Frank Gerth III Dissertation Award): *Gradient-based Optimization and implicit regularization over non-convex landscapes*